

Line Keep Assist (LKA) with Evaluation and Failure Analysis

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Abstract—This work presents a complete Line Keep Assist (LKA) perception pipeline based purely on classical computer vision. The system combines HLS color and Sobel gradient thresholding, inverse perspective mapping, histogram-based sliding windows, quadratic lane fitting and temporal smoothing to detect left and right lane boundaries from a monocular front camera, producing both an annotated video and a per-frame CSV log with detection flags, confidence scores and lateral offset. The method is evaluated on two clips: a daytime highway sequence and a low-light rural road. On the highway clip it achieves 97.44% left and 96.74% right detection accuracy with a lateral offset of 0.565 ± 0.671 m, while on the rural clip it reaches 99.54% accuracy on both sides with an offset of 0.047 ± 0.068 m. These results show that a tuned classical pipeline can deliver stable LKA perception in a limited operational design domain, but remains sensitive to curvature, approximate geometry and competing edges, motivating integration with calibrated and deep-learning-based approaches.

I. INTRODUCTION

Line Keep Assist (LKA) is an important component of modern Advanced Driver Assistance Systems (ADAS). Its role is to detect lane boundaries, estimate the vehicle's position within the lane, and identify situations where the driver may be drifting. A stable LKA system improves safety and driving comfort, especially on highways and long journeys.

This project implements a complete LKA perception pipeline using classical computer vision techniques. The approach relies on color and gradient thresholding, inverse perspective mapping, sliding-window search, polynomial curve fitting, and temporal smoothing. Each frame of the input video is processed to extract lane markings, compute confidence scores, and estimate the lateral offset of the car. The final output includes an annotated video with real-time overlays and a per-frame CSV log of all quantitative metrics.

Scope and Contributions:

- **Classical CV Pipeline:** A full lane-detection system designed with HLS thresholding, Sobel gradients, perspective warping, and quadratic fitting of left and right lane boundaries.
- **Temporal Smoothing and Confidence:** Stabilization of lane estimates over time using exponential filtering, combined with confidence scoring based on pixel count, fit residuals, and lane continuity.
- **Quantitative Evaluation:** Computation of side detection accuracy, mean lateral offset, stability, and flicker behavior directly from the generated CSV logs.
- **Qualitative Analysis:** Examination of success and failure cases across different road conditions such as shadows, weak paint, curves, and merges.

- **Implementation Assumptions:** Camera calibration was not performed, and the perspective transform relies on empirically selected source and destination points.

This setup provides a clear, interpretable baseline for lane detection and allows a detailed look at the strengths and limitations of classical image-processing methods in real driving scenarios.

II. METHOD (CLASSICAL CV BASELINE)

The pipeline processes each frame of the input video and extracts left and right lane boundaries using a sequence of classical image-processing steps. The method is fully deterministic and does not require training.

A. ROI and Perspective Warp (IPM)

Only the lower region of the frame is used for lane analysis. A homography is applied to map the road surface into a bird's-eye view. Using four manually selected source points, the transformation is:

$$p' = Hp, \quad (1)$$

where p is a pixel in the camera frame, p' is the corresponding point in top-down space, and H is the 3×3 homography matrix. Camera calibration was not used; the mapping was tuned empirically.

B. Preprocessing: HLS and Sobel

The frame is converted to HLS, and the S-channel is thresholded to capture bright lane paint. To emphasize vertical edges, a horizontal Sobel operator is applied on the L-channel:

$$G_x = \frac{\partial L}{\partial x} = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix} * L. \quad (2)$$

Pixels passing both the color and gradient thresholds form a binary mask representing potential lane markings.

C. Lane Pixel Extraction

In the warped binary image, a histogram of x-intensities over the lower half identifies the left and right lane bases. A sliding-window search is then used to gather pixels along each side. Windows shift horizontally using the mean x -location of detected pixels, while regions with insufficient support are ignored.

D. Curve Fit and Projection

For each lane side, the collected pixel coordinates (x_i, y_i) are fitted with a second-order polynomial:

$$x(y) = ay^2 + by + c. \quad (3)$$

Fits with unrealistic curvature are rejected. The resulting curves are projected back onto the camera frame using the inverse homography H^{-1} .

E. Temporal Smoothing

To avoid frame-to-frame jitter, polynomial coefficients are smoothed using an exponential moving average:

$$\theta_t = \alpha\theta_{t-1} + (1 - \alpha)\hat{\theta}_t, \quad (4)$$

where $\hat{\theta}_t$ are the coefficients from the current frame and α controls smoothness. If detection is weak, previous coefficients are reused for a limited number of frames.

F. Confidence and Detection Flags

Each lane side receives a confidence score based on pixel count, residual fit error, and vertical span:

$$C = w_1 \tanh\left(\frac{N}{k}\right) + w_2 e^{-E/\beta} + w_3 S, \quad (5)$$

where N is the number of pixels, E is the mean residual of the polynomial fit, and S is the detected vertical coverage. A lane is considered detected if $C > 0.6$.

G. HUD and CSV Outputs

The system annotates each processed frame with a compact HUD showing the left/right detection status, the corresponding confidence values, and the estimated lateral offset. This provides an intuitive summary of the lane state and allows quick visual debugging during playback.

In parallel, the pipeline logs one row per frame into a CSV file. The recorded fields are:

- `frame_id` — sequential frame index,
- `left_detected`, `right_detected` — binary lane flags,
- `left_conf`, `right_conf` — confidence scores in $[0, 1]$,
- `lat_offset_m` — lateral displacement in meters.

An excerpt from the generated CSV is shown in Fig. 1. It illustrates how confidences rise as the pipeline stabilizes and how the detection flags switch from 0 to 1 when both lanes become reliably visible.

III. EVALUATION METRICS

This work adopts several quantitative metrics commonly used in lane-keeping perception systems. These metrics evaluate geometric accuracy, temporal behavior, and detection reliability. All values are computed directly from the per-frame CSV logs.

frame_id	left_detected	right_detected	left_conf	right_conf	lat_offset_m
0	0	0	0.13743217175564829	0.13933678664179558	-0.12750303537703192
1	0	0	0.25424951774794935	0.2577722423910828	-0.1275117669682461
2	0	0	0.35403263001915697	0.3584250638582721	-0.12604453829793894
3	0	0	0.4388596314096813	0.4439827424899921	-0.12475727973240557
4	0	0	0.5108142671777798	0.5167476198731656	-0.12425756291544272
5	0	0	0.5718228896603	0.5786876544162274	-0.12405534926654527
6	1	1	0.6238093247728675	0.631338141526137	-0.12385097568937004
7	1	1	0.6681626134951735	0.676044203157383	-0.12433961744497918

Fig. 1. Excerpt from the per-frame CSV output, showing lane detection flags, confidence scores, and lateral offset.

A. Metric Definitions

- **Side Detection Accuracy:** proportion of frames where the algorithm outputs the correct YES/NO detection for each lane side.
- **Mean Confidence:** average confidence score per lane, combining pixel support, fit residual, and vertical span.
- **Lateral Offset:** estimated vehicle displacement from the lane center in meters at the bottom of the image. Reported using mean and standard deviation.
- **Temporal Stability:** mean absolute change in offset between consecutive frames:

$$S_{\Delta} = \frac{1}{N-1} \sum_{t=2}^N |o_t - o_{t-1}|$$

- **Flicker Rate:** fraction of frames in which a lane's detection flag toggles (YES $\leftrightarrow$ NO). Lower values indicate greater temporal consistency.

B. Summary of Metrics

Table I reports the computed metrics for two evaluation clips: a daytime highway sequence and a low-light rural sequence.

TABLE I
COMPUTED EVALUATION METRICS FOR BOTH TEST CLIPS.

Metric	Video 1	Video 2
Acc L (%)	97.44	99.54
Acc R (%)	96.74	99.54
Mean Off (m)	0.565	0.047
Off Std (m)	0.671	0.068
Temp Stab (m/f)	0.0391	0.0035
Flicker	0.002	0.001

IV. RESULTS

A. Quantitative

Table I summarizes the behaviour of the LKA pipeline on two clips: a straight highway sequence (Video 1) and a rural road sequence (Video 2). In both cases, side-detection accuracy is very high. On the highway clip the system reaches 97.44% left and 96.74% right accuracy, clearly above the 90%

target. On the rural clip the detector is almost perfect, with 99.54% accuracy for both sides.

The lateral offset statistics highlight the difference between the two scenarios. For Video 1 the mean offset is 0.565 m with a relatively large standard deviation of 0.671 m. This matches the temporal plot in Fig. 8, where the vehicle stays close to the right lane boundary for most of the clip and then undergoes a large excursion around frames 300–360. The histogram in Fig. 10 is bimodal, with modes near ≈ 0.9 m and ≈ -1.0 m, indicating that the tracker sometimes locks onto the wrong lane boundary when the road curves and the perspective warp becomes less accurate.

In contrast, Video 2 shows behaviour much closer to what an LKA system would ideally produce. The mean offset is only 0.047 m with a small standard deviation of 0.068 m. The temporal plot in Fig. 9 stays near zero with gentle drift, and the histogram in Fig. 11 is narrow and centred around zero. This indicates that the vehicle remains close to the lane centre for almost the entire clip.

Temporal stability and flicker metrics support the same trend. On the highway clip the mean absolute change in offset is 0.0391 m/frame with a flicker rate of 0.002 per frame, which is acceptable but shows some instability when the geometry changes. On the rural clip temporal stability improves to 0.0035 m/frame and the flicker rate halves to 0.001, meaning that lane labels almost never blink on and off once the system has locked on.

B. Qualitative Analysis

Figure 2 and Fig. 3 show the intermediate stages of the pipeline for one frame from each clip. For Video 1 the HLS+Sobel mask is noisy due to strong texture and shadows, and the bird’s-eye projection contains many vertical structures coming from guard rails and road markings. Even in this case the bottom-half histogram (Fig. 4) still exhibits clear peaks that allow the algorithm to initialise left and right lane bases. For Video 2 the binary mask and bird’s-eye view are much cleaner: the two lane lines appear as bright vertical bands with very little clutter, and the corresponding histogram (Fig. 5) has two sharp peaks.

Example HUD overlays are shown in Fig. 6 and Fig. 7. On the highway sequence, frames 50 and 150 have a lateral offset of about 0.9 m, meaning the ego vehicle is closer to the left guard rail within the detected ego-lane. By frame 350 the estimated offset flips to -0.74 m, which indicates that the tracker has switched to a different pair of lines (for example, confusing the inner dashed line with the outer shoulder). This behaviour explains the negative lobe in the offset histogram and is a typical failure mode of the classical sliding-window approach.

For the rural clip the three sample frames are much more consistent: offsets of -0.11 m, -0.01 m and 0.10 m (Fig. 7) show that the vehicle stays close to the lane centre. The HUD also reports high and stable confidence (≈ 0.9 – 1.0) for both sides.



Fig. 2. Video 1 (highway): original frame, HLS+Sobel threshold mask, and bird’s-eye warped binary.



Fig. 3. Video 2 (rural): original frame, HLS+Sobel threshold mask, and bird’s-eye warped binary.

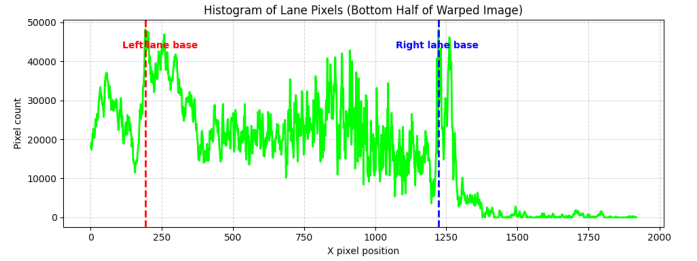


Fig. 4. Video 1: histogram of lane pixels in the bottom half of the warped binary image.

confidence curves remain near one and no large offset spikes appear.

The smoothed–offset curves in Fig. 12–13 illustrate the effect of temporal filtering. For Video 1 the rolling mean removes short-term noise but still clearly shows the large excursion when the road curvature changes; in a real LKA controller this would likely trigger a “driver takeover” warning. For Video 2 the smoothed offset is almost flat, confirming that the classical CV pipeline can be very stable when the lane paint is strong and the scene is less cluttered.

Overall, the results show that the classical HLS+Sobel + IPM + sliding-window baseline can reach high accuracy and low flicker on well-marked roads, but it remains sensitive to sudden geometry changes and to scenes where other strong edges compete with the true lane markings.

V. ODD AND FAILURE MODES

A. Operational Design Domain

The implemented LKA is evaluated in a narrow operational design domain (ODD):

- **Sensor:** single front-facing RGB camera, 30 fps.
- **Road type:** marked asphalt lanes with visible paint.
- **Conditions:** daytime and dusk, dry road, moderate curvature, limited traffic.
- **Test clips:**
 - Video 1 – highway driving with dashed and solid white lines.

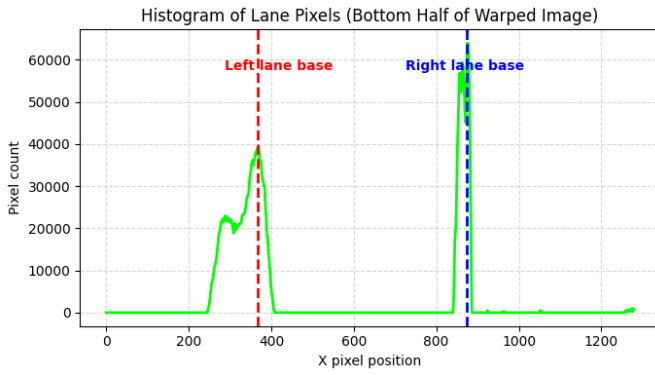


Fig. 5. Video 2: histogram of lane pixels in the bottom half of the warped binary image.

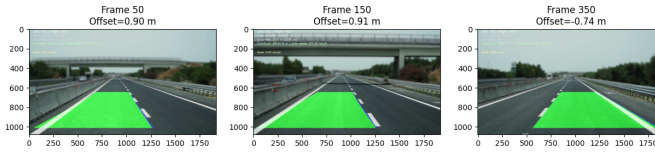


Fig. 6. Video 1: HUD overlays at frames 50, 150, and 350 showing strong lateral movement and large offset variation.

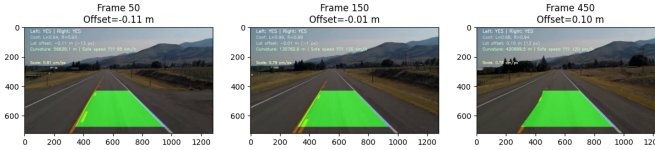


Fig. 7. Video 2: HUD overlays at frames 50, 150, and 450 illustrating stable detection and small offsets near lane center.

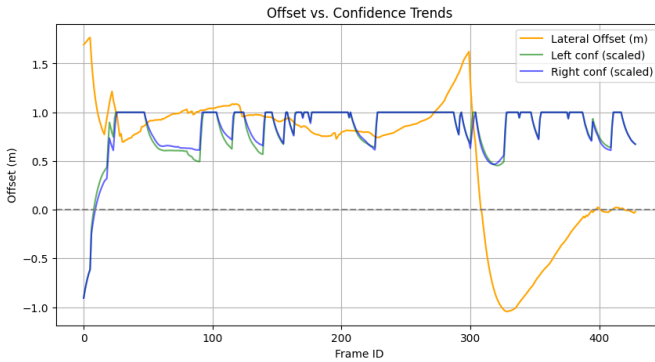


Fig. 8. Video 1: lateral offset and left/right confidence over time. Large offset drift and temporary confidence drops are visible on curves and merges.

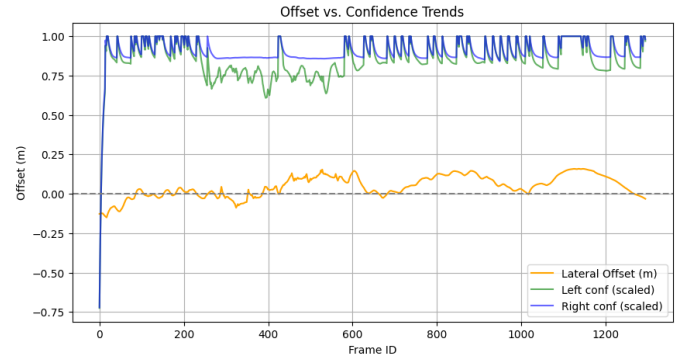


Fig. 9. Video 2: lateral offset and confidence over time. Offset remains close to zero with high temporal stability.

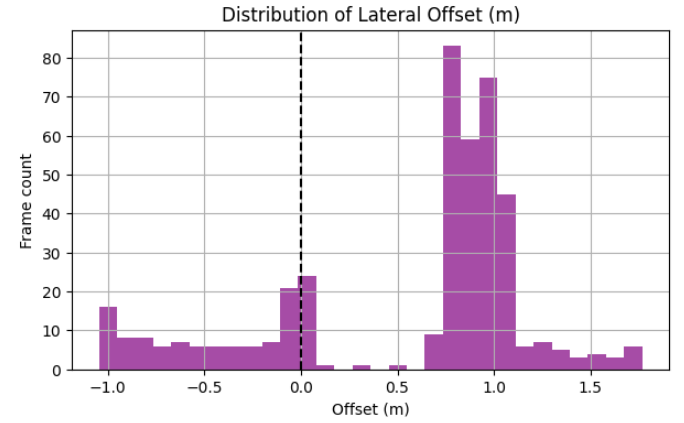


Fig. 10. Video 1: distribution of lateral offset values. The bimodal shape indicates lane-identity confusion under highway curvature and perspective.

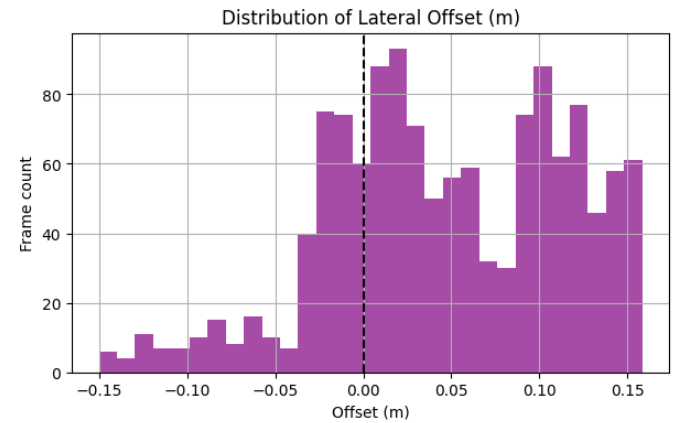


Fig. 11. Video 2: distribution of lateral offset values. Narrow, single-peaked distribution centred near zero indicates stable trajectory tracking.

- Video 2 – low-light rural road with yellow centre line and white edges.

Scenarios such as heavy rain, night-time glare, fog, snow, unmarked roads or construction zones are out-of-ODD for this baseline system.

B. Observed Failure Modes

Figure 10 and Figure 11 show that Video 1 has a wide, bimodal offset distribution, while Video 2 is narrow and centred near zero. This reflects several characteristic failure modes:



Fig. 12. Video 1: raw and smoothed lateral offset over time. Smoothing removes noise but still reflects large drift during curved-road segments.

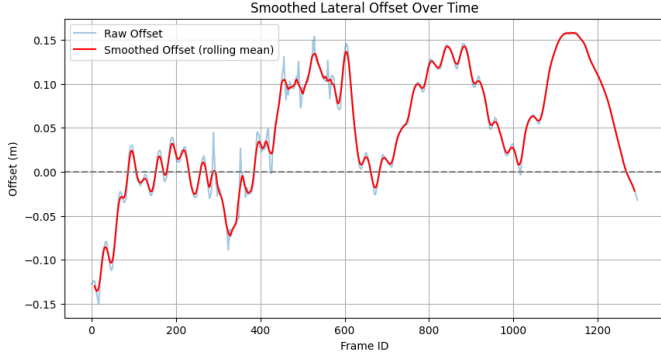


Fig. 13. Video 2: raw and smoothed lateral offset over time. Both curves remain close, showing high stability and low noise.

- **Lane identity confusion on curves (Video 1):** sliding windows occasionally jump from the true right boundary to the inner dashed line, causing large offset swings despite high confidence.
- **Offset bias and drift:** approximate IPM points and unknown camera centre introduce a systematic bias (about 0.5–1 m in Video 1), even when the lane is tracked.
- **Spurious edges:** guard rails and bright road texture are sometimes picked up by the HLS+Sobel mask and can pull the polynomial fit sideways.
- **Weak markings:** when dashed segments or worn paint reduce the pixel count, the lane can disappear for a few frames, although temporal smoothing keeps the flicker rate low.

Table II summarises the main failure types and the corresponding behaviour that a safety-oriented LKA should adopt.

VI. ABLATIONS AND ROBUSTIFIERS

The final configuration was obtained by tuning a small set of hyperparameters: S-channel thresholds [90, 255], Sobel-x range [15, 120], $N_{\text{windows}} = 9$, window margin = 110 px, MINPIX = 40, and EMA factor $\alpha = 0.9$ with a 12-frame memory.

TABLE II
OBSERVED FAILURE MODES AND SUGGESTED LKA RESPONSE.

Failure mode	Symptom in metrics	Suggested action
Lane ID confusion on curves	Large jumps in lateral offset, bimodal histogram (Video 1)	Freeze steering, warn driver, reduce assist torque
Systematic bias	Mean offset $\approx 0.5\text{--}1\text{ m}$, stable confidence	Use rate-of-change, limit corrections, log for calibration
Spurious clutter	edges / Short spikes in offset, local drops in confidence	Reject short vertical spans, down-weight low-confidence frames
Weak / missing markings	Conf. $< \tau$ for several frames, occasional flicker	After N low-conf. frames: disengage LKA, keep warning only

- **Thresholds:** adjusted HLS and Sobel ranges kept lane paint visible in both bright highway and low-light rural scenes, while limiting shadows and texture noise.
- **Window geometry:** a wider margin and lower MINPIX helped maintain continuous lanes on dashed and partially faded markings.
- **Temporal smoothing:** the stronger EMA is directly reflected in the low flicker rates (0.002 for Video 1 and 0.001 for Video 2), despite occasional noisy frames.

All metrics in Table I correspond to this final setting.

VII. CONCLUSION

This work implemented a classical LKA perception pipeline using HLS+Sobel thresholding, IPM, sliding-window lane tracking, quadratic fitting and temporal smoothing, producing an annotated video and a per-frame CSV log. On the highway clip (Video 1), the system reached 97.44% left and 96.74% right detection accuracy, with a mean lateral offset of 0.565 m and standard deviation 0.671 m. On the rural low-light clip (Video 2), accuracy rose to 99.54% on both sides, with a mean offset of 0.047 m and standard deviation 0.068 m, indicating much higher stability.

The comparison highlights a typical trade-off for classical CV: it works very well when markings are clear and contrast is high, but is sensitive to curvature, approximate homography and competing edges, which appear as offset drift and bimodal offset distribution on Video 1. Future improvements include adding camera calibration and metric scaling, testing contrast enhancement (e.g. CLAHE) for night and rain, using a Kalman filter for temporal fusion, and feeding this post-processing pipeline with a deep learning lane-segmentation model to extend the operational design domain.

ARTIFACTS

All code, notebooks, annotated videos and per-frame CSV logs used in this report are available at: <https://github.com/jim0002/Line-Keep-Assist-LKA->.